ŘEDITELSTVÍ SILNIC A DÁLNIC ČR

NDIC - DATEX II Elaborated Data Publication - Traffic Status

Release 1.1rev0

Národní Dopravní Informační Centrum

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—

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Author
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This Specification defines data format, used for publishing information about traffic status on predefined set of locations.

CHAPTER

ONE

INTRODUCTION

This document shall be used by publisher and / or subscriber. Subscriber shall use it for format suitability evaluation and for possible implementation of received NDIC data import into own information system. Publisher shall use it as a specification of data to be published format and for possible implementation of data export from own information system into NDIC.

Here you can find:

- conceptual description of published information
- information about location referencing methods used
- W3C XML schema of the format
- data sample(s)

Following topics are out of scope of this document and if present, shall be considered as informative only:

- transfer protocol incl. access point, life cycle etc.
- related data formats
- typical data origin
- typical data use

1.1 General terms

Following terms are used in this document:

format
: description of data structure for transferring some kind of information

protocol
: description of data transfer process and rules

NDIC
: national traffic information centre

DATEX II model
: class model for describing information related to the road traffic. The model in form of UML is platform independent.

  One of specific platforms for expressing the model is XML.

  The model is extensive (it consists of few thousands of structural items representing term incl. definition) For reference data (UML class model in form of Enterprise Architect EAP file, XML schema etc.) see [d2ref]. For additional information see [d2supp].

(DATEX II) profile
: simplified class model, still sufficient for specific needs. Profile is created from model by selecting only specific parts.

(DATEX II) extension
extended DATEX II model, adding data structures missing in standard model but needed for specific needs.

B level extension
Extension of DATEX II model, which is backward compatible with the original DATEX II model, even though it contains additional structures. [16157-1]

C level extension
Extension of DATEX II model, which is not backward compatible with the original DATEX II model.

publication
an element of DATEX II model, serving as an envelope for transmitted content of certain kind. Every sent DATEX II message contains exactly one element of publication type. There are various types of publications, for example *SituationPublication* [16157-3], *ElaboratedDataPublication* [16157-5], *PredefinedLocationsPublication* [16157-3] etc.

(W3C) XML schema
language, which allows definition of rules for a XML document structure.

XSD
XML document containing XML schema. Sometimes it is synonym for XML schema itself.

uri
uniform resource identifier, text, used as unique identifier. In this document, every format has it's own uri.

CHAPTER

TWO

CONCEPTS

Described format allows publishing traffic status information for set of predefined routes.

2.1 Publication type ElaboratedDataPublication

Format is based on DATEX II publication of type *ElaboratedDataPublication* [16157-5] and is used to publish information about traffic status on defined routes.


```xml
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel xmlns="http://datex2.eu/schema/2/2_0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" modelBaseVersion="2"
xsi:schemaLocation="http://datex2.eu/schema/2/2_0 ../schema/schema.xsd">
  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>NDIC</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication xsi:type="ElaboratedDataPublication" lang="cz">
    <feedType>x-format:cz-ndic_d2-traffic-status-v1.1</feedType>
    <publicationTime>2017-07-28T16:05:04+02:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>NDIC</nationalIdentifier>
    </publicationCreator>
    <timeDefault>2017-07-28T16:10:04+02:00</timeDefault>
    <headerInformation>
      <confidentiality>noRestriction</confidentiality>
      <informationStatus>real</informationStatus>
    </headerInformation>
    <elaboratedData>
      <source>
        <sourceIdentification>ndic-secid-1-01</sourceIdentification>
      </source>
```


Traffic status for particular route is given in element *elaboratedData*, which can appear in the publication multiple times, describing traffic statuses on distinct routes. Element *elaboratedData* does not have its own identifier. Its uniqueness is guaranteed by publication time and by identifier of referenced location.

2.2 Predefined locations on road network

The *route* is described by a linear section on the road network, given by the presence of a vehicle detection. Since the position of the vehicle detectors or sections where the traffic status is calculated does not change, the location in this profile is described by a reference (id) to a predefined location set defined in different document. The reference points to a class instance *PredefinedLocation* from package *PredefinedLocationsPublication*. Therefore, the location is not described directly in the file.

Route of status information (traffic status) is determined by reference to a set of predefined locations. To look up given location, it is necessary to obtain publication *PredefinedLocationsPublication*, related to this source of information and inside find location of identical type (by *targetClass*, here *PredefinedLocation*), with the same identifier and version.

The route for which the calculated traffic status is valid is specified by the *pertinentLocation* element. It always has *xsi:type="LocationByReference"* and includes element *predefinedLocationReference* with the attribute *targetClass="PredefinedLocation"*.

```
        <pertinentLocation xsi:type="LocationByReference">
          <predefinedLocationReference targetClass="PredefinedLocation"
          version="0" id="6D476A44-AA44-4947-AA4B-7C4DD79BB67E" />
        </pertinentLocation>
```

2.3 Traffic status (similar to Czech "traffic level")

The traffic status (density) is not expressed in levels (1-5), as it is customary in the Czech Republic, but by phrases, such as "Congestion". To allow expression in the DATEX II phrases the the converters are used.

```
        <trafficStatus>
          <trafficStatusValue>heavy</trafficStatusValue>
        </trafficStatus>
```

Traffic status is expressed by value from enumeration *TrafficStatusEnum* as defined in standard [16157-3] in table A.125.

Table 1: Values contained in "TrafficStatusEnum"

Enumerated type name	Purpose	Definition
impossible	Impossible	Traffic is fully congested in given direction, which makes driving impossible.
congested	Congested traffic	Traffic is congested in given direction and driving is slow and difficult.
heavy	Heavy	Traffic is heavier than usual in given direction and it makes driving more difficult than usual.
freeFlow	Free flow traffic	Traffic stream is fluent in given direction.
unknown	Unknown	Traffic conditions are unknown.

Other attributes, such as trend, are not used. In the Czech version of the profile the definitions are adhered to definitions of "czech" traffic levels [tl-wiki].

2.4 Publishing only currently available data

Format provides information about traffic status only for routes, for which data are available. Other routes, with no data available are omitted or value *unknown* is used.

CHAPTER

THREE

TIME VALIDITY

The start of validity of traffic statuses is given once for all records in the publication by the *timeDefault* element, of the type *DateTime* and class *ElaboratedDataPublication*.

The stated validity time represents the current time to which the calculated data relates, it is the end of the measurement period.

No further specification of validity is used.

```
    <feedType>x-format:cz-ndic_d2-traffic-status-v1.1</feedType>
    <publicationTime>2017-07-28T16:05:04+02:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>NDIC</nationalIdentifier>
    </publicationCreator>
    <timeDefault>2017-07-28T16:10:04+02:00</timeDefault>
```

CHAPTER FOUR

LOCATION REFERENCING

The format describes locations exclusively by reference to a set of predefined routes (class *predefinedLocation*) in publication *PredefinedLocationsPublication* [16157-3]..

Location referencing methods are out of scope of this document and are described elsewhere.

CHAPTER

FIVE

APPENDIX A: MESSAGE EXAMPLES

5.1 traffic-levels.xml - traffic status on two routes

Publication contains two records about traffic status for two different routes.

Routes are defined in different publication of type *PredefinedLocationsPublication*, which is not described here in detail.

First route, with identifier *6D476A44-AA44-4947-AA4B-7C4DD79BB67E*, has traffic status *heavy*, which means heavy traffic.

Second route, with identifier *5C8463E7-C547-4D30-B22A-4C95BEDE96B6*, has traffic status *congested*, which means congested traffic.


```xml
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel xmlns="http://datex2.eu/schema/2/2_0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" modelBaseVersion="2"
xsi:schemaLocation="http://datex2.eu/schema/2/2_0 ../schema/schema.xsd">
  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>NDIC</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication xsi:type="ElaboratedDataPublication" lang="cz">
    <feedType>x-format:cz-ndic_d2-traffic-status-v1.1</feedType>
    <publicationTime>2017-07-28T16:05:04+02:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>NDIC</nationalIdentifier>
    </publicationCreator>
    <timeDefault>2017-07-28T16:10:04+02:00</timeDefault>
    <headerInformation>
      <confidentiality>noRestriction</confidentiality>
      <informationStatus>real</informationStatus>
    </headerInformation>
    <elaboratedData>
      <source>
        <sourceIdentification>ndic-secid-1-01</sourceIdentification>
      </source>
      <basicData xsi:type="TrafficStatus">
        <pertinentLocation xsi:type="LocationByReference">
          <predefinedLocationReference targetClass="PredefinedLocation"
          version="0" id="6D476A44-AA44-4947-AA4B-7C4DD79BB67E" />
        </pertinentLocation>
        <trafficStatus>
```


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```xml
          <trafficStatusValue>heavy</trafficStatusValue>
        </trafficStatus>
      </basicData>
    </elaboratedData>
    <elaboratedData>
      <basicData xsi:type="TrafficStatus">
        <pertinentLocation xsi:type="LocationByReference">
          <predefinedLocationReference targetClass="PredefinedLocation"
          version="0" id="5C8463E7-C547-4D30-B22A-4C95BEDE96B6" />
        </pertinentLocation>
        <trafficStatus>
          <trafficStatusValue>congested</trafficStatusValue>
        </trafficStatus>
      </basicData>
    </elaboratedData>
  </payloadPublication>
</d2LogicalModel>
```


5.1.1 Root element and publication envelope

Root element *d2LogicalModel* declares, that this message is using DATEX II.

Element *exchange* contains provider identifier.

Element *payloadPublication* is an envelope for actual information payload containing general information and one or more elements *elaboratedData*, (two in our case).

Attribute *xsi:type="ElaboratedDataPublication"* in element *payloadPublication* determines, that this envelope is of type Elaborated data publication.

Element *feedType* states publication schema by its uri.

Element *publicationTime* gives time when publication has been created.

Element *publicationCreator* identifies creator of the publication.

timeDefault element gives start time of validity for all of calculated traffic statuses, the time, for which all traffic status values were calculated.

5.1.2 Description of traffic status at one place

Following element *elaboratedData* can be repeated and describes a calculated traffic status on a particular route.

Element *sourceIdentification* gives the identifier to the measuring site / equipment additionally to location identifier.

Nested element *basicData* states by it's attribute *xsi:type="TrafficStatus"*, that given values are of type "traffic status".

Traffic status location

Attributes *id ="6D476A44-AA44-4947-AA4B-7C4DD79BB67E"* and *id=" 5C8463E7-C547-4D30-B22A-4C95BEDE96B6"* in elements *predefinedLocationReference* of distinct records *elaboratedData* refers to two predefined locations (routes) with a given identifiers, and the attribute *version ='0'* specifies the version of the location description.

Traffic status

Element *trafficStatus* gives in *trafficStatusValue* traffic status by enumerated type value. In this case by value *heavy* for the first location and *congested* for the second location.

Time validity

Time validity (start time) of calculated traffic statuses is determined together for all records in element *timeDefault*.

CHAPTER

SIX

APPENDIX B: W3C XML SCHEMA

XML structure of a transmitted message is defined by following W3C XML schema:

6.1 *schema.xsd*

```xml
<?xml version="1.0" encoding="utf-8" standalone="no"?>
<xs:schema elementFormDefault="qualified" attributeFormDefault="unqualified" xmlns:D2LogicalModel="http://datex2.eu/schema/2/2_0" version="2.3" targetNamespace="http://datex2.eu/schema/2/2_0" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="_ExtensionType">
    <xs:sequence>
      <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs="unbounded" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="_PredefinedLocationVersionedReference">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:VersionedReference">
        <xs:attribute name="targetClass" use="required" fixed="PredefinedLocation" />
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="AreaOfInterestEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="continentWide" />
      <xs:enumeration value="national" />
      <xs:enumeration value="neighbouringCountries" />
      <xs:enumeration value="notSpecified" />
      <xs:enumeration value="regional" />
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="BasicData" abstract="true">
    <xs:sequence>
      <xs:element name="pertinentLocation" type="D2LogicalModel:GroupOfLocations" minOccurs="0" />
      <xs:element name="basicDataExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="Boolean">
    <xs:restriction base="xs:boolean" />
  </xs:simpleType>
  <xs:simpleType name="ComputationMethodEnum">
    <xs:restriction base="xs:string">
```

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```
      <xs:enumeration value="arithmeticAverageOfSamplesBasedOnAFixedNumberOfSamples" /
↪>
      <xs:enumeration value="arithmeticAverageOfSamplesInATimePeriod" />
      <xs:enumeration value="harmonicAverageOfSamplesInATimePeriod" />
      <xs:enumeration value="medianOfSamplesInATimePeriod" />
      <xs:enumeration value="movingAverageOfSamples" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="ConfidentialityValueEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="internalUse" />
      <xs:enumeration value="noRestriction" />
      <xs:enumeration value="restrictedToAuthorities" />
      <xs:enumeration value="restrictedToAuthoritiesAndTrafficOperators" />
      <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndPublishers" />
      <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndVms" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="CountryEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="at" />
      <xs:enumeration value="be" />
      <xs:enumeration value="bg" />
      <xs:enumeration value="ch" />
      <xs:enumeration value="cs" />
      <xs:enumeration value="cy" />
      <xs:enumeration value="cz" />
      <xs:enumeration value="de" />
      <xs:enumeration value="dk" />
      <xs:enumeration value="ee" />
      <xs:enumeration value="es" />
      <xs:enumeration value="fi" />
      <xs:enumeration value="fo" />
      <xs:enumeration value="fr" />
      <xs:enumeration value="gb" />
      <xs:enumeration value="gg" />
      <xs:enumeration value="gi" />
      <xs:enumeration value="gr" />
      <xs:enumeration value="hr" />
      <xs:enumeration value="hu" />
      <xs:enumeration value="ie" />
      <xs:enumeration value="im" />
      <xs:enumeration value="is" />
      <xs:enumeration value="it" />
      <xs:enumeration value="je" />
      <xs:enumeration value="li" />
      <xs:enumeration value="lt" />
      <xs:enumeration value="lu" />
      <xs:enumeration value="lv" />
      <xs:enumeration value="ma" />
      <xs:enumeration value="mc" />
      <xs:enumeration value="mk" />
      <xs:enumeration value="mt" />
      <xs:enumeration value="nl" />
      <xs:enumeration value="no" />
      <xs:enumeration value="pl" />
```

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```
      <xs:enumeration value="pt" />
      <xs:enumeration value="ro" />
      <xs:enumeration value="se" />
      <xs:enumeration value="si" />
      <xs:enumeration value="sk" />
      <xs:enumeration value="sm" />
      <xs:enumeration value="tr" />
      <xs:enumeration value="va" />
      <xs:enumeration value="other" />
    </xs:restriction>
  </xs:simpleType>
  <xs:element name="d2LogicalModel" type="D2LogicalModel:D2LogicalModel" />
  <xs:complexType name="D2LogicalModel">
    <xs:sequence>
      <xs:element name="exchange" type="D2LogicalModel:Exchange" />
      <xs:element name="payloadPublication" type="D2LogicalModel:PayloadPublication" minOccurs="0" />
      <xs:element name="d2LogicalModelExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
    <xs:attribute name="modelBaseVersion" use="required" fixed="2" />
  </xs:complexType>
  <xs:complexType name="DataValue" abstract="true">
    <xs:sequence>
      <xs:element name="dataError" type="D2LogicalModel:Boolean" minOccurs="0" maxOccurs="1" />
      <xs:element name="reasonForDataError" type="D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
      <xs:element name="dataValueExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
    <xs:attribute name="accuracy" type="D2LogicalModel:Percentage" use="optional" />
    <xs:attribute name="computationalMethod" type="D2LogicalModel:ComputationMethodEnum" use="optional" />
    <xs:attribute name="numberOfIncompleteInputs" type="D2LogicalModel:NonNegativeInteger" use="optional" />
    <xs:attribute name="numberOfInputValuesUsed" type="D2LogicalModel:NonNegativeInteger" use="optional" />
    <xs:attribute name="smoothingFactor" type="D2LogicalModel:Float" use="optional" />
    <xs:attribute name="standardDeviation" type="D2LogicalModel:Float" use="optional" />
    <xs:attribute name="supplierCalculatedDataQuality" type="D2LogicalModel:Percentage" use="optional" />
  </xs:complexType>
  <xs:simpleType name="DateTime">
    <xs:restriction base="xs:dateTime" />
  </xs:simpleType>
  <xs:complexType name="ElaboratedData">
    <xs:sequence>
      <xs:element name="source" type="D2LogicalModel:Source" minOccurs="0" />
      <xs:element name="basicData" type="D2LogicalModel:BasicData" minOccurs="0" />
      <xs:element name="elaboratedDataExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ElaboratedDataPublication">
```

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```
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:PayloadPublication">
        <xs:sequence>
          <xs:element name="periodDefault" type="D2LogicalModel:Seconds" minOccurs="0
↪" maxOccurs="1" />
          <xs:element name="timeDefault" type="D2LogicalModel:DateTime" minOccurs="0"␣
↪maxOccurs="1" />
          <xs:element name="headerInformation" type="D2LogicalModel:HeaderInformation
↪" />
          <xs:element name="elaboratedData" type="D2LogicalModel:ElaboratedData"␣
↪maxOccurs="unbounded" />
          <xs:element name="elaboratedDataPublicationExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="Exchange">
    <xs:sequence>
      <xs:element name="supplierIdentification" type=
↪"D2LogicalModel:InternationalIdentifier" />
      <xs:element name="exchangeExtension" type="D2LogicalModel:_ExtensionType"␣
↪minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="Float">
    <xs:restriction base="xs:float" />
  </xs:simpleType>
  <xs:complexType name="GroupOfLocations" abstract="true">
    <xs:sequence>
      <xs:element name="groupOfLocationsExtension" type="D2LogicalModel:_ExtensionType
↪" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="HeaderInformation">
    <xs:sequence>
      <xs:element name="areaOfInterest" type="D2LogicalModel:AreaOfInterestEnum"␣
↪minOccurs="0" maxOccurs="1" />
      <xs:element name="confidentiality" type="D2LogicalModel:ConfidentialityValueEnum
↪" minOccurs="1" maxOccurs="1" />
      <xs:element name="informationStatus" type="D2LogicalModel:InformationStatusEnum
↪" minOccurs="1" maxOccurs="1" />
      <xs:element name="urgency" type="D2LogicalModel:UrgencyEnum" minOccurs="0"␣
↪maxOccurs="1" />
      <xs:element name="headerInformationExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="InformationStatusEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="real" />
      <xs:enumeration value="securityExercise" />
      <xs:enumeration value="technicalExercise" />
      <xs:enumeration value="test" />
    </xs:restriction>
  </xs:simpleType>
```

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```
  <xs:complexType name="InternationalIdentifier">
    <xs:sequence>
      <xs:element name="country" type="D2LogicalModel:CountryEnum" minOccurs="1" maxOccurs="1" />
      <xs:element name="nationalIdentifier" type="D2LogicalModel:String" minOccurs="1" maxOccurs="1" />
      <xs:element name="internationalIdentifierExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="Language">
    <xs:restriction base="xs:language" />
  </xs:simpleType>
  <xs:complexType name="Location" abstract="true">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:GroupOfLocations">
        <xs:sequence>
          <xs:element name="locationExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="LocationByReference">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:Location">
        <xs:sequence>
          <xs:element name="predefinedLocationReference" type="D2LogicalModel:_PredefinedLocationVersionedReference" minOccurs="1" maxOccurs="1" />
          <xs:element name="locationByReferenceExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MultilingualString">
    <xs:sequence>
      <xs:element name="values">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="value" type="D2LogicalModel:MultilingualStringValue" maxOccurs="unbounded" />
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="MultilingualStringValue">
    <xs:simpleContent>
      <xs:extension base="D2LogicalModel:MultilingualStringValueType">
        <xs:attribute name="lang" type="xs:language" />
      </xs:extension>
    </xs:simpleContent>
  </xs:complexType>
  <xs:simpleType name="MultilingualStringValueType">
    <xs:restriction base="xs:string">
```

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```
      <xs:maxLength value="1024" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="NonNegativeInteger">
    <xs:restriction base="xs:nonNegativeInteger" />
  </xs:simpleType>
  <xs:complexType name="PayloadPublication" abstract="true">
    <xs:sequence>
      <xs:element name="feedType" type="D2LogicalModel:String" minOccurs="0" maxOccurs="1" />
      <xs:element name="publicationTime" type="D2LogicalModel:DateTime" minOccurs="1" maxOccurs="1" />
      <xs:element name="publicationCreator" type="D2LogicalModel:InternationalIdentifier" />
      <xs:element name="payloadPublicationExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
    <xs:attribute name="lang" type="D2LogicalModel:Language" use="required" />
  </xs:complexType>
  <xs:simpleType name="Percentage">
    <xs:restriction base="D2LogicalModel:Float" />
  </xs:simpleType>
  <xs:simpleType name="Seconds">
    <xs:restriction base="D2LogicalModel:Float" />
  </xs:simpleType>
  <xs:complexType name="Source">
    <xs:sequence>
      <xs:element name="sourceIdentification" type="D2LogicalModel:String" minOccurs="0" maxOccurs="1" />
      <xs:element name="sourceExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="String">
    <xs:restriction base="xs:string">
      <xs:maxLength value="1024" />
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="TrafficStatus">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:BasicData">
        <xs:sequence>
          <xs:element name="trafficStatus" type="D2LogicalModel:TrafficStatusValue" minOccurs="0" />
          <xs:element name="trafficStatusExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="TrafficStatusEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="impossible" />
      <xs:enumeration value="congested" />
      <xs:enumeration value="heavy" />
      <xs:enumeration value="freeFlow" />
```

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```
      <xs:enumeration value="unknown" />
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="TrafficStatusValue">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:DataValue">
        <xs:sequence>
          <xs:element name="trafficStatusValue" type="D2LogicalModel:TrafficStatusEnum" minOccurs="1" maxOccurs="1" />
          <xs:element name="trafficStatusValueExtension" type="D2LogicalModel:ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="UrgencyEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="extremelyUrgent" />
      <xs:enumeration value="urgent" />
      <xs:enumeration value="normalUrgency" />
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="VersionedReference">
    <xs:attribute name="id" type="xs:string" use="required" />
    <xs:attribute name="version" type="xs:string" use="required" />
  </xs:complexType>
</xs:schema>
```

CHAPTER

SEVEN

CHANGELOG

version 1.1.0

- added feed type to schema and samples
- added sourceIdentification to samples

version 1.0.1

- revision of the documentation, proofreading, changes to examples

version 1.0.0

- Initial format and documentation

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